

Q:- Interface program

```
import java.util.Scanner;
interface FixedDeposit {
    void getDetails();
    void showDetails();
    void calcInterest();
}

class HDFC implements FixedDeposit {
    String name, pan;
    int age;
    double amount, years, interest;

    public void getDetails() {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Name: ");
        name = sc.nextLine();

        System.out.print("Enter PAN: ");
        pan = sc.nextLine();

        System.out.print("Enter Age: ");
        age = sc.nextInt();

        System.out.print("Enter Deposit Amount: ");
        amount = sc.nextDouble();

        System.out.print("Enter Number of Years: ");
        years = sc.nextDouble();
    }

    public void calcInterest() {
        interest = amount * 0.04 * years;
    }

    public void showDetails() {
        System.out.println("HDFC FIXED DEPOSIT DETAILS");
        System.out.println("Name: " + name);
        System.out.println("PAN: " + pan);
    }
}
```

```

        System.out.println("Age: " + age);
        System.out.println("Interest Earned: Rs. " + interest);
    }
}

class IndianBank implements FixedDeposit {
    String name, aadhar, phone;
    double amount, years, interest;

    public void getDetails() {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Name: ");
        name = sc.nextLine();

        System.out.print("Enter Aadhar Number: ");
        aadhar = sc.nextLine();

        System.out.print("Enter Phone Number: ");
        phone = sc.nextLine();

        System.out.print("Enter Deposit Amount: ");
        amount = sc.nextDouble();

        System.out.print("Enter Number of Years: ");
        years = sc.nextDouble();
    }

    public void calcInterest() {
        interest = (amount * 0.045 * years) + (300 * years);
    }

    public void showDetails() {
        System.out.println("\nINDIAN BANK FIXED DEPOSIT DETAILS");
        System.out.println("Name: " + name);
        System.out.println("Aadhar: " + aadhar);
        System.out.println("Phone: " + phone);
        System.out.println("Interest Earned: Rs. " + interest);
    }
}

```

```

public class Run {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("Choose Bank for Fixed Deposit:");
        System.out.println("1. HDFC");
        System.out.println("2. Indian Bank");
        System.out.print("Enter your choice: ");
        int choice = sc.nextInt();

        FixedDeposit fd = null;

        if (choice == 1) {
            fd = new HDFC();
        } else if (choice == 2) {
            fd = new IndianBank();
        } else {
            System.out.println("Invalid Choice!");
            return;
        }

        fd.getDetails();
        fd.calcInterest();
        fd.showDetails();
    }
}

```

OUTPUT:-

```

PS C:\Users\student> & 'C:\Program Files\Java\jdk-24\bin\java.exe' '-agentlib:jdwp=transport=dt_socket,server=n,suspend=y,address=localhost:52144' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\student\AppData\Local\Temp\vscodesw_55e43\jdt_ws\jdt.ls-java-project\bin' 'Run'
Choose Bank for Fixed Deposit:
1. HDFC
2. Indian Bank
Enter your choice: 1
Enter Name: shahabaz
Enter PAN: 1234
Enter Age: 19
Enter Deposit Amount: 3000000
Enter Number of Years: 20
HDFC FIXED DEPOSIT DETAILS
Name: shahabaz
PAN: 1234
Age: 19
Interest Earned: Rs. 2400000.0
PS C:\Users\student> & 'C:\Program Files\Java\jdk-24\bin\java.exe' '-agentlib:jdwp=transport=dt_socket,server=n,suspend=y,address=localhost:58597' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\student\AppData\Local\Temp\vscodesw_55e43\jdt_ws\jdt.ls-java-project\bin' 'Run'
Choose Bank for Fixed Deposit:
1. HDFC
2. Indian Bank
Enter your choice: 2
Enter Name: Mohad
Enter Aadhar Number: 8123424472
Enter Phone Number: 45678943743
Enter Deposit Amount: 4444
Enter Number of Years: 20
INDIAN BANK FIXED DEPOSIT DETAILS
Name: Mohad
Aadhar: 8123424472
Phone: 45678943743
Interest Earned: Rs. 9999.6
PS C:\Users\student>

```